BAI070 Principles of Generative AI

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to		
CO 1	Understand the fundamental concepts and real-world applications of Artificial Intelligence and Generative AI tools and platforms.	K_2
CO 2	Apply prompt engineering techniques to design optimized and context-aware prompts for generative tasks in various AI platforms.	K_3
CO 3	Analyze the effectiveness of generative AI in domains such as education, office automation, marketing, and healthcare through specific use cases.	K_4
CO 4	Evaluate and compare various generative AI tools for content generation, code assistance, media creation, and automation.	K_5
CO5	Design AI-based workflows for real-world applications using tools like ChatGPT, GitHub Copilot, Jasper, and others	K_5
DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Foundations and Fundamentals of Generative AI:** Introduction to Artificial Intelligence: Definitions, Foundations, Scope, and Applications, Implications of AI across industries, Fundamentals of Generative AI: Concept, Evolution, Capabilities, Applications of Generative AI in various domains, Introduction to Generative AI platforms: ChatGPT (OpenAI), Gemini (Google), GitHub Copilot (Microsoft)	**08**
II	**Prompt Engineering and Prompt Optimization Techniques:**Basics of Prompt Engineering, Types of Prompts: Declarative, Interrogative, Instructional, Advanced Prompt Engineering: Chain-of-Thought Prompting, Zero-shot and Few-shot Prompting, Prompt Optimization Techniques, Use Cases and Practical Examples in Coding, Writing, and Customer Interaction	**08**
III	**Generative AI Applications in Workplace and Education:** AI in Office Productivity: Email Writing, Report Drafting, Minutes of Meeting (MoM), Proposal Writing, Summarization, and Task Automation:, Generative AI in Education: AI for Content Creation, Summarization, and Explanation, Curriculum Design using AI Tools, AI in Assessment, Feedback, and Visualization (e.g., Mindmaps, Infographics)	**08**
IV	**Domain-Specific Applications of Generative AI:** Generative AI in Digital Marketing: Content Strategy for YouTube, Facebook, Instagram, X (Twitter), LinkedIn and AI in Professional Branding, Email Marketing Automation and AI in E-Commerce; Generative AI in Healthcare and Agriculture: Diagnosis Assistance, Patient Data Summarisation, Crop Monitoring, Yield Prediction using AI	**08**
V	**Creative and Technical Aspects of Generative AI:** AI-Assisted Programming (e.g., Code Generation, Auto Debugging using Copilot, GPT-4), Generative AI in Creative Domains: Image & Design Generation (DALL·E, Midjourney), Music Composition with AI Tools; AI in Gaming & Animation (e.g., Scenario Generation, NPC Scripting), Customer Interaction Automation: Chatbots, Virtual Assistants, and Voice AI; Exploration of Popular AI Tools & Platforms: RunwayML, Jasper, Synthesia, Lumen5, Bard, etc.	**08**

Text Book (s)

1. Emerson, J, *Ripples of generative AI: Navigating AI's impact on society*. AI Press (2023).

Reference Book (s)

1. Pandit, R., *Generative AI* (Hardcover), Notion Press (2024).
2. OpenAI. (n.d.). *OpenAI*, from https://openai.com/
3. Google. (n.d.). *Gemini*, from https://gemini.google.com/
4. Microsoft. (n.d.). *Copilot*, from https://copilot.microsoft.com/chats/VkLP7TbPcVtu4dDjpDVTv

SWAYAM/NPTEL/MOOCs Certification:

1. https://onlinecourses.swayam2.ac.in/imb25_mg135/preview
2. https://onlinecourses.swayam2.ac.in/nou25_ge85/preview